# THERMAL REVISION DECK

## PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS



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CARD #01 THERMODYNAMICS

### NEWTON'S LAW OF COOLING

**Question:** What fundamental equation governs the rate of heat loss of a hot core capsule submerged in a cooler ambient fluid?

FRONT: QUESTION SPHERE STUDY DECK

CARD #01 KEY FORMULA

### EXPONENTIAL DECAY

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

**Explanation:** The cooling rate $\frac{dT}{dt}$ is directly proportional to the instantaneous temperature gap $(T - T_{\text{env}})$.

BACK: REVEAL & LAW k = cooling constant

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CARD #02 THERMODYNAMICS

### DECAY CONSTANT (k)

**Question:** What does the constant $k$ quantify physically, and which capsule is better insulated: $k = 0.15\ \text{min}^{-1}$ or $k = 0.015\ \text{min}^{-1}$?

FRONT: QUESTION SPHERE STUDY DECK

CARD #02 INSULATION FACTOR

### LOWER-k CAPSULE (k = 0.015)

**Answer:** For a lumped system, $k = \frac{hA}{mc}$ is an effective cooling constant. Under comparable conditions, a smaller $k$ means a slower approach to the ambient temperature.

BACK: REVEAL & LAW Units: $\text{min}^{-1}$ or $\text{s}^{-1}$

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CARD #03 HEAT TRANSFER

### FOURIER'S CONDUCTION LAW

**Question:** How does thermal energy transfer directly through solid matter, and what formula defines conduction heat flow?

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CARD #03 CONDUCTION EQUATION

### MOLECULAR VIBRATION

$$\dot{Q} = -\lambda A \frac{dT}{dx}$$

**Explanation:** The conductive heat-transfer rate depends on thermal conductivity $\lambda$, area $A$ and the temperature gradient. Low-conductivity materials reduce the rate.

BACK: REVEAL & LAW $\lambda$ = thermal conductivity

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CARD #04 EQUILIBRIUM

### THERMAL EQUILIBRIUM

**Question:** When does net heat transfer between the capsule core and the $0^\circ\text{C}$ ice bath drop to zero? What physical law governs this?

FRONT: QUESTION SPHERE STUDY DECK

CARD #04 ZEROTH LAW

### EQUAL TEMPERATURES

**Answer:** When $T_{\text{capsule}} = T_{\text{env}}$. For the nominal ice bath this limit is $0^\circ\text{C}$, but the measured bath temperature should be used. With no temperature difference, there is no net heat transfer between them.

BACK: REVEAL & LAW Net heat-transfer rate = 0

SPHERE MISSION CONTROL

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## PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION (MLI)

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CARD #05 — RADIATION

### SPACE VACUUM RADIATION

**Question:** Which heat-transfer mechanism can cross a vacuum between separated bodies?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #05 — STEFAN-BOLTZMANN

### PHOTON HEAT FLOW

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_{\text{env}}^4)$$

**Answer:** Thermal radiation can cross a vacuum as electromagnetic energy. Conduction can still occur through physical contacts within a spacecraft, but not across an empty gap.

BACK: REVEAL & LAW — $\sigma = 5.67\times10^{-8}\ \mathrm{W/m^2K^4}$

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CARD #06 — SHIELDING

### MYLAR RADIATIVE SHIELD

**Question:** How does a thin shiny aluminized Mylar foil protect spacecraft and astronauts from extreme solar thermal radiation?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #06 — OPTICAL REFLECTION

### INFRARED REFLECTANCE

**Answer:** An aluminized reflective film can have high infrared reflectance and low emissivity. Its performance depends on the specific product, surface condition and whether it faces a suitable gap; it is not a substitute for conductive insulation in direct water contact.

BACK: REVEAL & LAW — Verify product optical properties

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CARD #07 — CONVECTION

### CONVECTIVE AIR TRAPPING

**Question:** How do the sealed air cells in bubble wrap reduce heat transfer?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #07 — STAGNANT CELLS

### STAGNANT AIR POCKETS

**Answer:** Air has low thermal conductivity ($\lambda_{\text{air}} \approx 0.026\ \mathrm{W/(m\cdotp K)}$), while small sealed cells suppress bulk air circulation.

BACK: REVEAL & LAW — Low gas conduction + suppressed circulation

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CARD #08 — MLI SPACESUITS

### MULTI-LAYER INSULATION

**Question:** Why do spacesuits and satellites stack Cotton + Bubble Wrap + Mylar in composite layers?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #08 — TRIPLE SHIELD

### TRIPLE HEAT SHIELD

**Answer: Composite synergy:**

- **Cotton:** Blocks solid conduction.
- **Bubble Wrap:** Traps fluid convection.
- **Mylar:** Reflects infrared radiation.

BACK: REVEAL & LAW — All 3 heat paths blocked

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## PART 3: REFERENCE MODEL, MEASUREMENTS & EXPERIMENTAL BOUNDARY

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CARD #09 — SIMULATION

### ANALYTICAL REFERENCE MODEL

**Question:** What assumptions does the SPHERE Newton cooling model make?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #09 — MODEL ASSUMPTIONS

### LUMPED-PARAMETER MODEL

**Answer:** It assumes a uniform capsule temperature, a constant bath temperature and a constant effective cooling coefficient. The model is a reference calculation, not a full digital twin.

BACK: REVEAL & LAW — Hardware + Simulation

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CARD #10 — LAB ANALOG

### ICE-WATER BOUNDARY

**Question:** Why is an ice-water slurry used in the comparative experiment?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #10 — CONSTANT SINK

### CONSTANT HEAT SINK

**Answer:** The slurry provides a relatively stable, repeatable cold boundary. Its actual temperature must be measured; it does not reproduce vacuum heat transfer.

BACK: REVEAL & LAW — $T_{\text{env}} = 0.0°\text{C}$ constant

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CARD #11 — THERMAL MASS

### HEAT CAPACITY OF WATER

**Question:** Why is liquid water ($c = 4184\ \text{J/kg}\cdot^\circ\text{C}$) ideal for capsule core telemetry trials? What is the heat formula?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #11 — THERMAL ENERGY

### HIGH THERMAL MASS

$$Q = m \cdot c \cdot \Delta T$$

**Answer:** Water stores large thermal energy per gram, producing smooth, measurable exponential decay over $15+\ \text{minutes}$ without dropping instantly.

BACK: REVEAL & LAW — $c = 4184\ \text{J/kg}\cdot\text{K}$

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CARD #12 — GRAPH ANALYSIS

### DECAY CURVE CURVATURE

**Question:** Why is the temperature-time graph $T(t)$ steep at $t = 0$ and increasingly flat toward $t = 15\ \text{min}$?

FRONT: QUESTION — SPHERE STUDY DECK

CARD #12 — ASYMPTOTIC FLOW

### SHRINKING TEMP GAP

**Answer:** Heat flux rate $\frac{dQ}{dt} \propto (T - T_{\text{env}})$. As the capsule cools, $(T - T_{\text{env}})$ shrinks, causing the rate of temperature drop to slow down asymptotically.

BACK: REVEAL & LAW — $\text{Slope} = -k(T - T_{\text{env}})$

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## PART 4: EXPERIMENTAL METHOD, ERROR METRICS & SYNTHESIS



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CARD #13 METRICS

### MEAN ABSOLUTE ERROR

**Question:** How is the mean absolute error between measured and predicted temperatures calculated?

FRONT: QUESTION SPHERE STUDY DECK

CARD #13 MODEL ACCURACY

### MODEL RESIDUALS

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |T_{\mathrm{meas},i} - T_{\mathrm{model},i}|$$

**Answer:** MAE is the mean magnitude of the residuals and is reported in $^\circ\mathrm{C}$. A lower value indicates closer agreement but does not, by itself, validate every model assumption.

BACK: REVEAL & LAW Units: °C

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CARD #14 LAB PROTOCOL

### PROBE PLACEMENT

**Question:** Why should the thermometer probe be positioned near the centre of the liquid rather than touching the capsule wall?

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CARD #14 BOUNDARY LAYER

### BULK FLUID TEMPERATURE

**Answer:** Wall contact biases the reading towards the local boundary temperature rather than the bulk liquid temperature and introduces a systematic measurement error.

BACK: REVEAL & LAW Use consistent probe position

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CARD #15 SI UNITS

### SI UNITS REFERENCE

**Question:** What are the standard SI units for temperature $T$, thermal energy $Q$, thermal conductivity $\lambda$, and cooling constant $k$?

FRONT: QUESTION SPHERE STUDY DECK

CARD #15 STANDARD UNITS

### UNITS CHEAT SHEET

- **Temperature ($T$):** Kelvin $(\mathrm{K})$ or Celsius $(^\circ\mathrm{C})$
- **Thermal Energy ($Q$):** Joules $(\mathrm{J})$
- **Decay Constant ($k$):** $\mathrm{s}^{-1}$ or $\mathrm{min}^{-1}$
- **Conductivity ($\lambda$):** $\mathrm{W}/(\mathrm{m}\cdot\mathrm{K})$

BACK: REVEAL & LAW Standard Metric Units

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CARD #16 DESIGN COMPARISON

### MULTILAYER CONFIGURATION

**Question:** Which tested configuration is assigned the smallest reference cooling constant, and which mechanisms may explain that choice?

FRONT: QUESTION SPHERE STUDY DECK

CARD #16 REFERENCE RESULT

### MULTILAYER TEST ASSEMBLY

**Answer:** The cotton, bubble-wrap and reflective-film assembly is assigned $k = 0.015\ \mathrm{min}^{-1}$. This is a model assumption to be evaluated against measurements, not a guaranteed material property.

BACK: REVEAL & LAW Compare with measured data